

Lineare Algebra
Serie 23
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Exercise 1

S is an orthonormal set $\Rightarrow S$ is linearly independent and $\|s_i\| = 1 \quad \forall s_i \in S$

$$S = \{s_1, s_2, \dots, s_n\} \quad \text{linear independent}$$

Extend S such that it forms a basis of V

$$B = \{s_1, \dots, s_n, v_1, \dots, v_m\}$$
$$\text{span}(B) = V$$

This basis is not yet orthonormal. With Gram-Schmidt we can transform every basis into an orthonormal basis.

$$\Rightarrow B' = \{s_1, \dots, s_n, v_{1'}, \dots, v_{m'}\}$$

Exercise 2

(a)

Let $A, C \in M_{n \times n}(\mathbb{R})$ with $C = AA^T$

$$c_{ij} = \sum_{k=1}^n a_{ik}(a_{kj})^T = \sum_{k=1}^n a_{ik}a_{jk} = \sum_{k=1}^n a_{jk}(a_{ki})^T = c_{ji}$$

□

(b)

Assume $A^T A = AA^T$ is invertible

$$x^T A^T A x = (Ax)^T (Ax) > 0$$
$$\Rightarrow Ax \neq 0 \quad \forall x \neq 0 \Rightarrow \ker(A) = \{0\}$$
$$\Rightarrow x^T A^T A x > 0$$

$\Rightarrow A$ positive definite

$A^T A$ positive definite. Assume A is not invertible then

$$\exists x \neq 0 : Ax = 0$$
$$\Rightarrow 0 = \|Ax\|^2 = (Ax)^T (Ax) = x^T A^T A x > 0$$

$\Rightarrow$ this is a contradiction

$\Rightarrow A$ is invertible.

□

(c)

Previous series

(d)

$$\langle Av_1, v_2 \rangle = (Av_1)^T v_2 = v_1^T A^T v_2 = \langle v_1, A^T v_2 \rangle = \langle v_1, Av_2 \rangle$$
$$\Rightarrow \langle Av_1, v_2 \rangle - \langle v_1, Av_2 \rangle = \lambda_1 \langle v_1, v_2 \rangle - \lambda_2 \langle v_1, v_2 \rangle = 0$$

since $\lambda_1 \neq \lambda_2$

$$\begin{aligned}\langle v_1, v_2 \rangle &= 0 \\ \Rightarrow v_1 &\perp v_2\end{aligned}$$

□

Exercise 3

$$A = QR = (v_1, v_2, v_3) = \begin{pmatrix} -1 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} \text{ Using Gram-Schmidt process}$$

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{v_1}{3} = \begin{pmatrix} \frac{-1}{3} \\ \frac{2}{3} \\ \frac{-2}{3} \end{pmatrix}$$

$$q_2 = v_2 - \langle v_2, q_1 \rangle q_1 = v_2 - (-1) \cdot q_1 = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}$$

$$q_3 = v_3 - \langle v_3, q_1 \rangle q_1 - \langle v_3, q_2 \rangle q_2 = v_3 - 1 \cdot q_1 - 0 \cdot q_2 = \begin{pmatrix} \frac{-2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$$

$$\Rightarrow A = QR = \begin{pmatrix} -1 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{-1}{3} & \frac{2}{3} & \frac{-2}{3} \\ \frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{-2}{3} & \frac{1}{3} & \frac{2}{3} \end{pmatrix} \cdot \begin{pmatrix} 3 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{use to solve for } Ax = b \text{ for } b = \begin{pmatrix} 0 \\ 3 \\ -3 \end{pmatrix}$$

$$Ax = QRx = b$$

$$\Rightarrow x = R^{-1}Qb$$

$$R^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{-1}{3} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Rightarrow x = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Exercise 4

(a)

Gram-Schmidt

to find a basis $B = \{b_1, b_2\}$ basis for U

$$q_1 = \frac{v_1}{\|v_1\|} = \frac{\sqrt{3}}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$q_2 = v_2 - \langle v_2, q_1 \rangle q_1 = v_2 - \sqrt{3} \cdot q_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow b_1 = q_1 = \frac{\sqrt{3}}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad b_2 = \frac{q_2}{\|q_2\|} = \begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \\ 0 \end{pmatrix}$$

$$\dim(U^\perp) = 3 - \dim(U) = 1$$

$$B' = \{b'_1\}$$

$$b'_1 = |b_1 \times b_2| = \frac{\sqrt{6}}{6} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \frac{\sqrt{6}}{6} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$$

(b)

$$\begin{aligned} \varphi : \mathbb{R}^3 &\rightarrow U \\ x &\mapsto Px \end{aligned}$$

with

$$P = b_1 b_1^T + b_2 b_2^T = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \frac{1}{6} \begin{pmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{pmatrix}$$

$$\begin{aligned} \psi : \mathbb{R}^3 &\rightarrow U^\perp \\ x &\mapsto P^\perp x \end{aligned}$$

with

$$P^\perp = b'_1 b_1'^T = \frac{1}{6} \begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ -2 & -2 & 4 \end{pmatrix} = I - P$$

Exercise 5

(a)

$$\Rightarrow U^\perp = \left\{ (a_n)_{n=0}^\infty \in V \mid \exists N \geq 0 : \forall m < N : a_m = 0 \right\}$$

since

$$\begin{aligned} \lim_{N \rightarrow \infty} N &= \infty \\ \Rightarrow U^\perp &= \left\{ (a_n)_{n=0}^\infty \in V \mid \forall m < \infty : a_m = 0 \right\} = \{0\} \end{aligned}$$

(b)

$$g \in U^\perp \quad f \in U$$

$$\Rightarrow \langle f, g \rangle = \int_0^1 f(x)g(x)dx = \int_0^{\frac{1}{2}} f(x)g(x)dx = \int_{\frac{1}{2}}^1 f(x)g(x)dx = 0$$

between $[\frac{1}{2}, 1]$ we have no information about the function $f(x)$, therefore

$$g(x) = \begin{cases} g(x) & x \in \mathbb{R} \setminus (\frac{1}{2}, 1) \\ 0 & x \in [\frac{1}{2}, 1] \end{cases}$$

One could give a long proof and show that it has to be constant on $[\frac{1}{2}, 1]$

$$\Rightarrow U^\perp = \left\{ g \in V \mid \forall x \in \left(\frac{1}{2}, 1\right) : g'(x) = 0 \right\}$$

Exercise 6

(a)

$$\begin{aligned} \delta : V &\rightarrow V^*, \quad v \mapsto \delta(v) = \langle v, \cdot \rangle \\ \delta(v)(w) &= \langle v, w \rangle \end{aligned}$$

δ is an isometry. Therefore

$$\langle \delta(v), \delta(w) \rangle^* = \langle v, w \rangle \quad \forall v, w \in V$$

since δ is an isomorphism

$$\begin{aligned} \forall v \in V \quad \exists! \varphi \in V^* : \delta(v) = \varphi \\ \forall w \in V \quad \exists! \psi \in V^* : \delta(w) = \psi \\ \Rightarrow \langle \varphi, \psi \rangle^* = \langle \delta(v), \delta(w) \rangle^* = \langle v, w \rangle \end{aligned}$$

Assume by contradiction there exists 2 scalar products on V^*

$$\begin{aligned} \langle \varphi, \psi \rangle_1^* &= \langle v, w \rangle \\ \langle \varphi, \psi \rangle_2^* &= \langle v, w \rangle \\ \Rightarrow \langle \varphi, \psi \rangle_1^* &= \langle \varphi, \psi \rangle_2^* \end{aligned}$$

they are equal

□

(b)

Define the Gram Matrices

$$\begin{aligned} G &= [\langle \cdot, \cdot \rangle]_B \\ G^* &= [\langle \cdot, \cdot \rangle^*]_{B^*} \end{aligned}$$

Let $\varphi, \psi \in V^* \quad v, w \in V : \quad \varphi = \delta^{-1}(v), \psi = \delta^{-1}(w)$

$$\begin{aligned} \langle \varphi, \psi \rangle^* &= \langle \delta^{-1}(\varphi), \delta^{-1}(\psi) \rangle = \langle v, w \rangle \\ G^* &= [\langle \cdot, \cdot \rangle^*]_{B^*} = ([\langle \cdot, \cdot \rangle]_B)^{-1} = G^{-1} \end{aligned}$$

$\Rightarrow$ The representation matrix of $\langle \cdot, \cdot \rangle^*$ with respect to B^* is equal to the inverse of the representation matrix of $\langle \cdot, \cdot \rangle$ with respect to B .



Single Coice

1. (d)
2. (a)
3. (d)
4. (b)

Multiple Choice Questions

1.
 - (a) true
 - (b) false
 - (c) false
 - (d) true
2.
 - (a) false
 - (b) true
 - (c) false
 - (d) true
3.
 - (a) false
 - (b) true
 - (c) true
 - (d) true